

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Some Types of Operating Systems



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features.

Understands the concepts of server, its characteristics and its relation with operating systems.

Can Do Statements:

- I can talk about operating systems.
- I can describe the simple batch operating system.
- I can explain the multi programmed operating system.

Ask your partner:

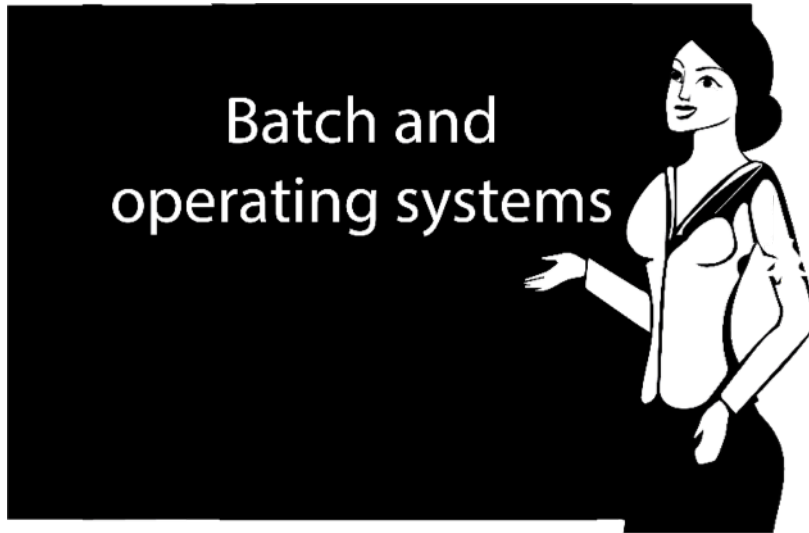
- How does a computer process information?
- What are the key components that made computers faster?





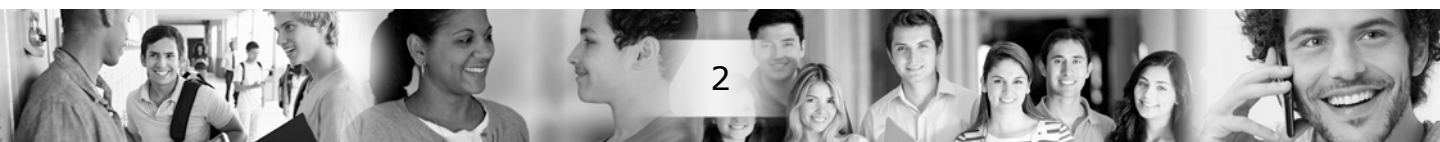
Let's Explore!

Look at a student presenting operating systems.



Instructions: Read the text below.

- Laura:** Today, I'm going to talk about batch and operating systems. However, if you want to learn about them, you will have to make sure you understand what an operating system is.
- Mark:** Yes, an operating system (OS) is the software that keeps a computer or device working. If you know how to handle and program them, you will have the opportunity to set up other drivers and applications.
- Laura:** That's a good definition of an OS. Now, as not all users have the same needs, there are many types of OS in the market with different characteristics. This leads us to the batch operating system. This one used to be the way in which lots of information was processed in early operating systems. In simple words, they get a command and start processing automatically all the information. If you want to check this OS, you will have to keep in mind its three main parts: a control card interpreter, which reads and carries out the instructions; a loader, which loads programs and commands to the memory; and, driver devices, which give the special set-ups to each system.
- Mark:** If a batch OS processes only one command at a time, will it mean that computers were slower, right?
- Laura:** Correct. That's where I wanted to get at. A lot of time was wasted waiting for a computer to end up a process before starting a new one. Luckily, we improved on that aspect and we launched the multiprogramming OS. This one analyzed the CPU activity and executed parts of programs simultaneously in the processor, giving the user the perception that they all are being executed at the same time. In other words, the computer is capable of cutting off a program for some time, executing another one, and then going after other tasks.





Let's Practice!

Let's talk over* operating systems!

* "Talk over" means "to discuss".

Instructions: Select all the correct answers.

Which of the following statements are linked up to Operating Systems?

- An operating system (OS) is the software that turns off a computer or device.
- An OS allows the setting up processes of drivers and applications according to the users' needs.
- An OS is the most important part of a system as it runs on the computer.
- With the OS we can look into the different prices of a system.
- The OS also keeps up with the specific needs of the user, allowing him to do different types of tasks.
- With the OS we can get at different operational levels in which we can operate different elements like the keyboard.



Let's Practice!

Let's describe the simple batch operating system!

Instructions: Write the words to correctly complete the sentences. Use words from the box below. Use: keep up / leave out / leads us to / carries out / sets up.

- The fact that customers have different needs _____ different operating systems, such as the batch operating system.
- The batch operating system allowed us to _____ with the tasks that we performed in early operating systems.
- You cannot _____ the fact that there are three main components of this operating system.
- The control card interpreter _____ the different instructions, while the loader loads programs and commands.
- Finally, the driver devices _____ each system.



Extension Activity

Let's write about multi programmed and simple batch Operating Systems!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about multi programmed and simple batch Operating Systems.
- Identify the main features of multi programmed and simple batch Operating Systems.
- Make a brochure describing what multi programmed and simple batch Operating Systems are and mention their main features.
- Use the grammar and the vocabulary related to this lesson.
- Share your brochure with your class.

Self-Assessment

Yes No

- I look for more information about multi programmed and simple batch Operating Systems.
- I identify the main features of multi programmed and simple batch Operating Systems.
- I make a brochure describing what multi programmed and simple batch Operating Systems are and mention their main features.
- I use the grammar and the vocabulary related to this lesson and I share my brochure with my class.

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Target Words

- **batch** (noun): a group of jobs or data taken together.
e.g. This leads us to the batch operating system.
- **capable** (adjective): having the ability of doing something.
e.g. In other words, the computer is capable of cutting off a program for some time, executing another one, and then going after other tasks.
- **CPU** (abbreviation): Central Process Unit, the part of a computer where the main instructions take place.
e.g. This one analyzed the CPU activity and executed parts of programs simultaneously in the processor, giving the user the perception that they all are being executed at the same time.

- **cut off** (phrasal verb): to cut or stop something.
e.g. In other words, the computer is capable of cutting off a program for some time, executing another one, and then going after other tasks.
- **lead** (verb): to guide or show the way.
e.g. This leads us to the batch operating system.
- **wasted** (adjective): used for no profit or benefit.
e.g. A lot of time was wasted waiting for a computer to end up a process before starting a new one.



Learning Points

Phrasal Verbs

Phrasal verbs are two-word verbs made out of a verb and a preposition. It is not easy to identify their meaning at first glance as it is not literal.

A **phrasal verb** is a combination of a verb plus a preposition, which together have a different meaning than the words separated. There are some important phrasal verbs that we have to learn:

PHRASAL VERB	MEANING
Catch up	to become equal or reach the same level
Come across	to find
Drop by	to visit in an unexpected way
Drop out	to stop attending a class
Get at	to reach something
Get through	to survive
Get up	to stand up
Go after	to try to stop someone; to try to catch something
Keep up with	to stay at the same level
Look into	to investigate
Put up with	to tolerate
Run into	to find or meet someone
Show up	to appear
Improve on /upon	to make something better

Examples

- *Through the OS it is possible to set up other drivers and applications.*
- *Before, a lot of time was wasted waiting for a computer to end up a process before starting a new one.*

Batch and Multi-Programmed Operating Systems

There are different types of operating systems, depending on the needs of the user.

A **batch operating system** is an operating system for users who need to process large amounts of information. It is an automated system which does not need manual intervention to avoid mistakes and speeds up the processing information.





Learning Points

Remember, a *batch* is a task which is done over and over again, so the batch operating system is programmed with specific commands, and then it starts processing all information without any other intervention. Here, we can be talking about millions of documents which need to be processed with the same command, which will be tedious and prone to mistakes if processed manually.

Multi-programmed operating systems were introduced to solve the problem of main memory in computers. Before, computers had to wait for a process to finish before starting a new one. Multiprogramming aims to execute multiple tasks at the same time while multiple batches are being processed. This type of technology has provided a more dynamic approach to computers, prioritizing processes and tasks and improving efficiency of times.

